Katam Akhilesh

+91 84639-15216 | Gmail | Linkedin | Github | Leetcode| Portfolio

Summary

Computer Science undergraduate skilled in AI, Generative AI, LLMs, RAG, Computer Vision, Full-Stack De velopment, React, Node.js, Django, Python, and IoT. Experienced in building and deploying scalable AI and Backend applications, chatbots, automation tools, and edge AI systems.

Technical Skills

Languages: Java, C/C++, Python, HTML, CSS, JavaScript, Typescript
Web Development: React, Node.js, Express, MongoDB, MySQL, Postgresql, REST APIs, UI/UX
Machine Learning and AI: YOLO, Model Deployment, OpenCV, EDA, Pandas, NumPy, scikit-learn, Seaborn, Matplotlib, SLR, LLMs, RAG
Developer Tools: Git, Postman, VS Code, PyCharm, Jupyter, Docker

Experience

AI + Software Engineering Intern Aug 2025 – Feb 2026
LOAM AI *Mumbai, India*

- Built AI chatbots, AI Enabled Features and RAG pipelines using LLMs with Langchain and LlamaIndex for enterprise applications with 20K+ active traffic and maintianed Software Engineering principles.
- Developed scalable REST APIs and backend services development using Django and PostgreSQL.
- Automated deployments with Docker, AWS based environment, CI/CD pipelines, and cloud based Applications.

Full Stack Development Intern May 2025 – Jul 2025
The Branding Jester *Hyderabad, Telangana*

- Worked on Building and integrating REST APIs using Node.js and Express.js for dynamic backend functionality
- Optimized performance, resolved production issues, maintained full-stack applications with User-centered design.

Projects

Digital Scarecrow | *Python, YOLOv8, OpenCV, Computer Vision, Raspberry Pi 5, IoT, Twilio, Blynk*

- Designed and deployed a real-time computer vision pipeline using YOLOv8n with PyTorch and OpenCV, achieving 94.2 mAP on a custom-labeled agricultural dataset of 2,000+ annotated images.
- Optimized model inference for edge deployment on Raspberry Pi 5, reducing detection latency to under 200ms through model quantization and frame-skipping strategies.
- Engineered an IoT dashboard with Blynk for real-time monitoring of sensor data, and detection events.

Justice AI | *Python, Flask, NLP, Machine Learning, Deep Learning, LLMs, RAG, Transformers, SQL*

- Developed an AI-powered Efficient Website with investigation platform for FIR analysis, intelligence extraction.
- Built NLP and RAG pipelines processing 10,000+ legal documents with sub-second FAISS retrieval for case summarization entity recognition along Vector Database
- Fine-tuned LLM-based chatbot via hyperparameter optimization for legal information retrieval and built a face recognition pipeline for suspect identification.

Achievements and Certifications

- 5x National Level Hackathon Winner — *HackerRank, TechSprint, Utkarsh, vCube Solutions, Webathon*
- 4x Coding Contest Winner — *GeeksforGeeks, OOPS! Code it Right, C Programming*
- Machine Learning Specialization — Andrew Ng, Coursera (2024)
- Google GenAI Study Jam — Google Developer Student Club

Patent

IoT-Based Crop Protection System with AI-Driven Animal Detection and Water Management

- Patent Published, 2025 (Application Number: 202541034995).

Education

Aditya Institute Of Technology and Management Srikakulam, Andhra Pradesh
Bachelor of Computer Science and Engineering - GPA : 8.79 *Aug. 2023 – May 2027*